

Seonghyeon Nam

PH.D., COMPUTER SCIENCE

EXPERTISE

Generative AI – Diffusion-/flow-based models for image/video generation, large-scale data curation, mid-/post-training of foundation models, SFT/GRPO, human-/VLM-based evaluation
Computer Vision, Machine Learning – 3D/4D view synthesis/stylization, computational photography, image processing
AI-native workflow – Data curation, coding/implementation, auto experiment/ablation, document/report generation

EXPERIENCE

Meta (Facebook), Menlo Park, California, United States

Senior Research Scientist

Feb' 22 - Present

- Pioneered research and development of CV/ML, diffusion-based image/video generation models for products in *Facebook*, *Meta Superintelligence Labs*, and *Meta Reality Labs*.
- **Facebook (Current):**
 - Built end-to-end pipeline of image-to-image (I2I) diffusion model post-training for human-centric image personalization including large-scale data curation, SFT/GRPO training, VLM-based autoeval.
 - Achieved production-level quality in the I2I post-training in terms of human identity preservation, prompt following, photorealism.
 - Accelerated a cycle of diffusion model training 4 times by adopting AI-native workflow.
- **Meta Superintelligence Labs:**
 - Delivered image-to-video (I2V) LoRA models to generate videos of visual effects.
 - Core-contributed to audio-conditioned video generation model for lipsync feature (speaking and singing faces) in Vibes app – several millions requests/day.
- **Meta Reality Labs:**
 - Developed on-device ML models for image/video processing (super-resolution, frame interpolation, etc) for AR/VR cameras (Quest, Ray-Ban Meta).
 - Developed image super-resolution model for camera and OCR/QR (AR glasses, Ray-Ban Meta).
 - Worked on 3D view synthesis/style transfer using NeRFs and 4D generative view synthesis.

York University, Toronto, Ontario, Canada

Postdoctoral Fellow

Jan' 21 - Jan' 22

- Worked on implicit neural representations for multi-image fusion and restoration.

Samsung AI Center, Toronto, Ontario, Canada

Postdoctoral Intern

Aug' 21 - Nov' 21

- Developed ML algorithm to recover RAW image from sRGB image for mobile cameras.

Snap Inc., Venice, California, United States

Research Intern

May' 18 - Aug' 18

- Developed deep generative model to synthesize time-lapse videos from a single image.

Yonsei University, Seoul, South Korea

Research Assistant

Mar' 14 - Aug' 20

- Worked on deep generative models for image/video synthesis.
 - Worked on computational photography and low-level vision.
-

EDUCATION

Yonsei University, Seoul, South Korea

Ph.D., Computer Science,

Mar' 14 - Aug' 20

Advisor: Prof. Seon Joo Kim

GPA: 4.10/4.3

PUBLICATIONS

M. Xie, N. Khan, T. Wang, N. Dhingra, **S. Nam**, H. Yang, Z. Hui, C. Metzler, A. Vedaldi, H. Pirsiavash, and L. Luo. LaVR: Latent Space Conditioned Video Re-rendering using Large 4D Reconstruction Models. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 2026.

H. Jung, **S. Nam**, N. Sarafianos, S. Yoo, A. Sorkine-Hornung, and R. Ranjan. Geometry Transfer for Stylizing Radiance Fields. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 2024.

Z. Wan, C. Richardt, A. Bozic, C. Li, V. Rengarajan, **S. Nam**, X. Xiang, T. Li, B. Zhu, R. Ranjan, and J. Liao. Learning Neural Duplex Radiance Fields for Real-Time View Synthesis. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

S. Yang, S. Jeon, **S. Nam**, and S. J. Kim. Dense Interspecies Face Embedding. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

S. Nam, M. A. Brubaker, and M. S. Brown. Neural Image Representations for Multi-Image Fusion and Layer Separation. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2022.

Y. H. Kim, **S. Nam**, and S. J. Kim. 2PESNet: Towards Online Processing of Temporal Action Localization. *Pattern Recognition (PR)* 131 (2022): 108871.

S. Nam, A. Punnappurath, M. A. Brubaker and M. S. Brown. Learning sRGB-to-Raw-RGB De-rendering with Content-Aware Metadata. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

D. Kim, J. W. Kim, **S. Nam**, D. Lee, Y. Lee, N. Kang, H.-E. Lee, B. Yoo, J.-J. Han, and S. J. Kim. Large Scale Multi-Illuminant (LSMI) Dataset for Developing White Balance Algorithm under Mixed Illumination. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2021.

Y. H. Kim, **S. Nam**, and S. J. Kim. Temporally Smooth Online Action Detection using Cycle-consistent Future Anticipation. *Pattern Recognition (PR)* 116 (2021): 107954.

S. Jeon, **S. Nam**, S. W. Oh, and S. J. Kim. Cross-Identity Motion Transfer for Arbitrary Objects through Pose-Attentive Video Reassembling. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2020.

Y. Kim, **S. Nam**, I. Cho, and S. J. Kim. Unsupervised Keypoint Learning for Guiding Class-Conditional Video Prediction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

S. Nam, C. Ma, M. Chai, W. Brendel, N. Xu, and S. J. Kim. End-to-End Time-Lapse Video Synthesis from a Single Outdoor Image. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

S. Nam, Y. Kim, and S. J. Kim. Text-Adaptive Generative Adversarial Networks: Manipulating Images with Natural Language. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018 (Spotlight).

S. Nam and S. J. Kim. Modelling the Scene Dependent Imaging in Cameras with a Deep Neural Network. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.

S. Nam^{*1}, Y. Hwang*, Y. Matsushita, and S. J. Kim. A Holistic Approach to Cross-Channel Image Noise Modeling and its Application to Image Denoising. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016 (**Spotlight**).

PATENT

Application

Methods, Apparatuses And Computer Program Products For Facilitating Motion Transfer For Stylizing Radiance Fields. **US Patent No. 19/411,115**

Registration

Method and Apparatus for Generating Video Based on Keypoints. **Korea Patent No. 10-2231391**

Apparatus and method for generating manipulated image based on natural language and system using the same. **Korea Patent No. 10-2192015**

Method and apparatus for image adjustment based on semantics-aware. **Korea Patent No. 10-2192016**

ACADEMIC SERVICE

Conference Reviewer

IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**)

IEEE International Conference on Computer Vision (**ICCV**)

European Conference on Computer Vision (**ECCV**)

Advances in Neural Information Processing Systems (**NeurIPS**)

International Conference on Learning Representations (**ICLR**)

AAAI Conference on Artificial Intelligence (**AAAI**)

Asian Conference on Computer Vision (**ACCV**)

Winter Conference on Applications of Computer Vision (**WACV**)

Journal Reviewer

IEEE Transactions on Pattern Recognition and Machine Intelligence (**TPAMI**)

IEEE Transactions on Image Processing (**TIP**)

Computer Vision and Image Understanding (**CVIU**)

HONORS & AWARDS

Outstanding Reviewer, ICCV 2021 2021

VISTA Postdoctoral Fellowship, \$55,000CAD/year, York University 2021

Postdoctoral Fellowship, \$39,000/year, National Research Foundation of Korea 2021

NAVER Fellowship, \$4,300, NAVER Corp. 2017

Excellent Paper Award, Dept. of Computer Science, Yonsei University 2016

Bronze Prize, \$4,300, 22nd Samsung HumanTech Paper Award 2016

Global Ph.D. Fellowship, \$26,000/year, National Research Foundation of Korea 2015 - 2019

SKILLS

Python, Pytorch, SQL, C/C++, MATLAB, HTML

¹Equal contribution